

EXECUTIVE SUMMARY

Innovation Proposal

Enhancing Kenya Power's Prepaid Electricity Experience
Through Bluetooth-Enabled Smart Customer Interface Unit (CIU)
Integration

Prepared by:

Kahindi Padii Gandhi

Android & Embedded Systems Developer

KCA University

Executive Summary

Kenya Power has made significant progress in improving customer experience through digital initiatives such as the **MyPower mobile application** and the **Tuende Digital** campaign. Customers can conveniently purchase prepaid electricity tokens using mobile payment platforms and access various digital services without visiting Kenya Power offices.

Despite these advancements, the final stage of the prepaid electricity process still requires customers to manually enter a 20-digit token into the Customer Interface Unit (CIU). In addition, customers must physically access the CIU to monitor remaining electricity units and other meter information. These limitations can lead to unexpected power interruptions, inconvenience, and reduced engagement with Kenya Power's digital services.

This proposal presents a **Bluetooth-enabled enhancement** to the existing prepaid electricity system. The proposed solution introduces a secure communication module integrated with the Customer Interface Unit, enabling the official MyPower mobile application to communicate directly with the CIU. Through this integration, customers would be able to securely transfer purchased prepaid tokens to the CIU, monitor electricity information from their smartphones, and receive low-balance notifications while preserving the existing prepaid infrastructure and manual token entry process.

A proof-of-concept prototype has been successfully developed to demonstrate the technical feasibility of the proposed solution. The prototype validates secure Bluetooth communication, real-time information exchange, and automated token transfer within a controlled environment, providing a practical foundation for future evaluation and integration.

The proposed solution offers several potential benefits to Kenya Power, including enhanced customer convenience, improved accessibility, increased adoption of the

MyPower mobile application, support for the **Tuende Digital** initiative, and a scalable platform for future smart energy services. Importantly, the proposal is designed as a **backward-compatible enhancement**, minimizing disruption to the current prepaid electricity infrastructure while providing opportunities for future innovation.

This proposal is respectfully submitted for technical evaluation and consideration as a potential enhancement to Kenya Power's ongoing digital transformation efforts. The objective is to explore opportunities for collaboration, prototype evaluation, and future integration with Kenya Power's existing prepaid electricity ecosystem.

Proposal Objective

The objective of this proposal is to demonstrate a practical, customer-focused enhancement to Kenya Power's prepaid electricity ecosystem that improves convenience, accessibility, and digital engagement while maintaining compatibility with existing prepaid infrastructure.

Request for Collaboration

I respectfully request an opportunity to present this proof-of-concept prototype to the appropriate Kenya Power technical and innovation teams for evaluation. I welcome the opportunity to discuss the proposal, receive professional feedback, and explore the feasibility of a pilot implementation or further collaborative development.

Contact Information

Kahindi Padii Gandi

Android & Embedded Systems Developer

Email: padiiamamea0@gmail.com

Phone: 0706900903